

```

syms t0 w;
t=0:0.001:1000;
f1=sin(3*pi*(t-2))./(pi*(t-2));
f1(t==2)=3;
F1=fft(f1);
f=0:0.001:1000;
magF1=abs(F1/1000);
phaseF1=unwrap(angle(F1));
f1_s=sin(3*pi*(t0-2))./(pi*(t0-2));
F1_s=fourier(f1_s,t0,w);
pretty(F1_s);

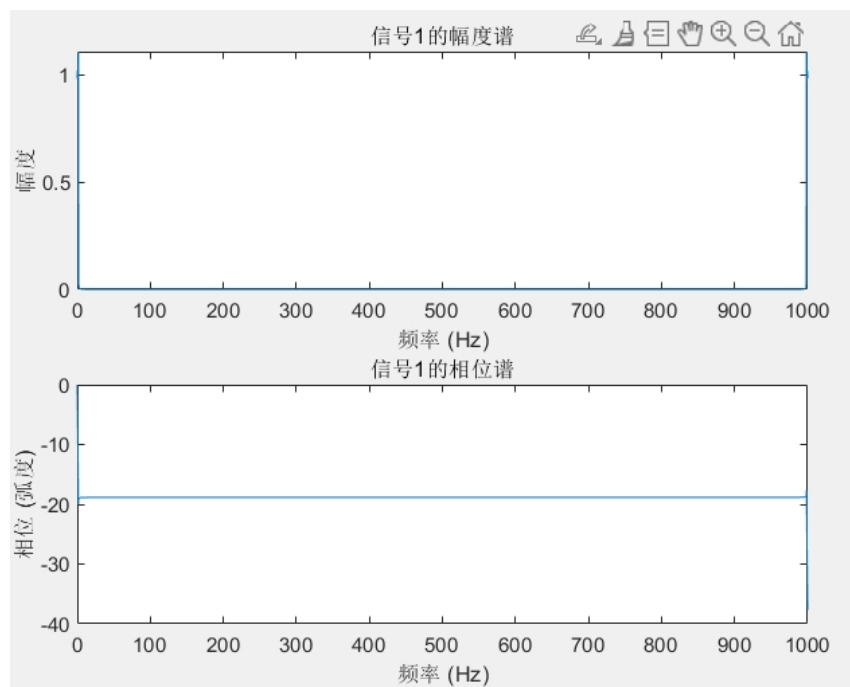
figure;
subplot(2,1,1);
plot(f, magF1);
title('信号1的幅度谱');
subplot(2,1,2);
plot(f, phaseF1);
title('信号1的相位谱');

f2=(sin(pi*t))./(pi*t);
f2(t==0)=1;
f2=f2.^2;
F2=fft(f2);
magF2 = abs(F2/1000);
phaseF2 = unwrap(angle(F2));
f2_s=((sin(pi*t0))./(pi*t0)).^2;
F2_s=fourier(f2_s,t0,w);
pretty(F2_s);

figure;
subplot(2,1,1);
plot(f, magF2);
title('信号2的幅度谱');
subplot(2,1,2);
plot(f, phaseF2);
title('信号2的相位谱');

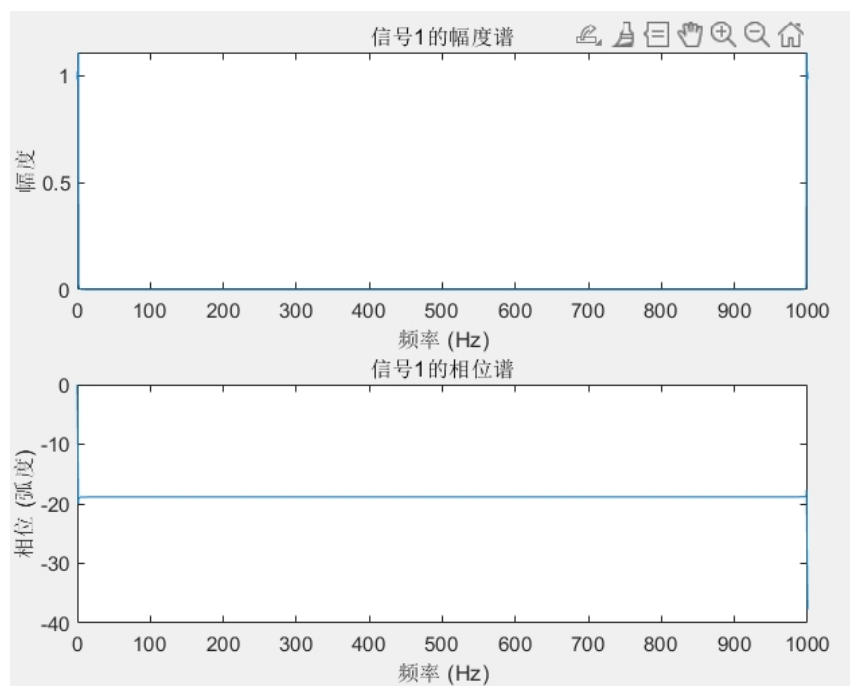
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输出(1)



$$\frac{\pi \operatorname{heaviside}(-w - 3\pi) \exp(-w 2i) - \pi \operatorname{heaviside}(3\pi - w) \exp(-w 2i)}{\pi}$$

(2)



$$\frac{\operatorname{fourier}\left(\frac{\cos(2\pi t_0)}{\sqrt{t_0^2 + \pi^2 w^2 \operatorname{sign}(w)}}, t_0, w\right)}{\pi}$$

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syms w t w0;
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x1=heaviside(w+w0)-heaviside(w-w0);
x2=dirac(w+w0)-dirac(w-w0);
x3=5*cos(2*w);
x4=(heaviside(w-1)-heaviside(w+1))*exp(-1i*w);
x1=ifourier(x1,w,t);
x2=ifourier(x2,w,t);
x3=ifourier(x3,w,t);
x4=ifourier(x4,w,t);
disp('x1的傅里叶反变换为:');
pretty(x1);
disp('x2的傅里叶反变换为:');
pretty(x1);
disp('x3的傅里叶反变换为:');
pretty(x1);
disp('x4的傅里叶反变换为:');
pretty(x1);

```

(1)输出为

x1的傅里叶反变换为:

$$\frac{\sin(t w_0) + \cos(t w_0) 1i}{2 \pi} - \frac{\sin(t w_0) + \cos(t w_0) 1i}{2 \pi}$$

(2)输出为

x2的傅里叶反变换为:

$$\frac{\exp(-t w_0 1i) - \exp(t w_0 1i)}{2 \pi}$$

(3)输出为

x3的傅里叶反变换为:

$$\frac{5 \operatorname{dirac}(t - 2)}{2} + \frac{5 \operatorname{dirac}(t + 2)}{2}$$

(4)输出为

x4的傅里叶反变换为:

$$\frac{-\sin(t - 1) + \cos(t - 1) 1i}{2 \pi} - \frac{\sin(t - 1) + \cos(t - 1) 1i}{2 \pi}$$